

HASIBUL HASAN

Senior Software Engineer • Agentic AI & Backend Systems

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PROFESSIONAL EXPERIENCE

Senior Software Engineer — Brain Station 23

2025 Jul – Present | Dhaka, Bangladesh

- Leading AI-driven development work, building agentic systems and workflows with **LangChain**, **LangGraph**, and **DeepAgent**.
- Designing and building multi-agent orchestration systems, async streaming AI chat backends, and **MCP**-based tool integrations for security operations tooling.
- Driving architectural decisions for AI agent platforms, mentoring on agentic design patterns, and maintaining high coding standards through regular code reviews.
- Built a real-time voice bot integrating **LLMs** with **Asterisk** (telephony/SIP) and **LiveKit** (WebRTC media streaming) to enable natural, low-latency voice conversations.

Tech Stack: Python, LangChain, LangGraph, DeepAgent, FastAPI, MCP, PostgreSQL, Redis, AWS (EC2), Docker, Asterisk, LiveKit

Software Engineer — Brain Station 23

2024 Jan – 2025 Jul | Dhaka, Bangladesh

- Actively contributed to the design, development, and deployment of scalable systems and applications using **Node.js**, **React.js**, **Serverless** Framework, and **AWS**.
- Working with **OpenAI** API, prompt engineering, MidJourney, and Flux to develop AI-driven features and solutions.
- Facilitated open agile workflow, sprint, and task management.
- Regular code reviews are conducted to maintain high coding standards and best practices.
- Proactive adoption and implementation of emerging technologies for enhanced system capability and alignment with industry trends.

Tech Stack: Python, JavaScript, TypeScript, React, Next.js, Express, Nest.js, Node.js, MongoDB, AWS Lambda, AWS S3, Amazon EC2

Associate Software Engineer — Brain Station 23

2022 May – 2023 Jan | Dhaka, Bangladesh

- Designed large-scale e-commerce platform that sustains above 100 users in a day focusing on optimization site performance and reducing first loading time using server-side rendering and ISR in **Next.js**.
- Developed a platform to socialize fitness freaks with a serverless Lambda function created in **Python** to process images and videos asynchronously. This transition from synchronous backend processing results in a 30% better performance.

Tech Stack: React.js, Next.js, Node.js, MongoDB, PostgreSQL, AWS

Software Engineer Trainee — Brain Station 23

2022 Jan – 2022 Apr | Dhaka, Bangladesh

- Built up a reality where the phone can capture and simulate all furniture and decoration items in a room, showcasing some of the best innovations found in augmented reality.
- Built a CV Builder application. With this project, I learned much about agility in the development process and improved my team collaboration skills.

Tech Stack: React.js, Next.js, Node.js, MongoDB, Three.js

SKILLS

Programming Languages: C/C++, Python, JavaScript, TypeScript

AI / Agentic Frameworks: LangChain, LangGraph, DeepAgent, MCP (Model Context Protocol), OpenAI API, RAG, Prompt Engineering

Voice / Telephony: LiveKit (WebRTC), Asterisk (SIP), Web Speech API

AWS Services: AWS Lambda, AWS Cognito, AWS S3, Amazon EC2, Elastic Beanstalk, CodePipeline, CodeBuild

Tools: Git, Jira, VSCode, Docker

Frameworks and Libraries: React, Next.js, Express, Nest.js, FastAPI, Serverless Framework

Databases: MySQL, Redis, MongoDB, PostgreSQL

EDUCATION

B.Sc. in Computer Science and Engineering

Daffodil International University | 2018 Jan – 2022 Jan | Dhaka, Bangladesh

PROJECTS

Anomali — AI-Driven Security Operations Platform (ThreatStream & Agent Platform) (Current)

Project Duration: 2024 Jan – 2026 Sep

Core engineering work at Brain Station 23 on Anomali's **ThreatStream** and Agent Platform product line, building an AI-driven SOC pipeline across the full alert-to-case lifecycle. Alerts surfaced by detection and alert rules flow into the system, where AI agents analyze and triage

each one to confirm whether it's a true or false positive; confirmed alerts escalate into incidents, and related incidents are grouped into cases with automatically generated investigation reports. The goal throughout has been to compress the repetitive, manual parts of a SOC analyst's day — reading logs, cross-referencing threat intel, writing up findings — into an agentic pipeline that reasons over the same data an analyst would, while surfacing enough evidence at each step for a human analyst to validate the pipeline's conclusions through human-in-the-loop review.

Tech Stack: Python · FastAPI · LangGraph · LangChain · MCP (Model Context Protocol) · PostgreSQL · Redis · WebSockets · AWS · Docker · ReportLab

- Designed the end-to-end alert → incident → case pipeline: AI agents validate whether triaged alerts are true or false positives, escalate confirmed ones into incidents, and generate case-level reports summarizing grouped incidents.
- Built the AI pipeline that verifies the alert → incident → case workflow behaves as expected end-to-end, generating investigation analysis in the same style a human SOC analyst would — backed by supporting evidence at each stage — and routing findings through human-in-the-loop (HITL) approval before a case is finalized.
- Built the multi-agent orchestration system on **LangGraph** — **WebSocket** ingress, orchestrator routing, and parallel ReAct subagents (Threat Intel, Payload, Identity, Enterprise Context) — on a **FastAPI** backend with a **PostgreSQL** relational schema, async **SQLAlchemy** ORM, and real-time **WebSocket** streaming (resolved **AsyncPostgresSaver**/astream architecture issues).
- Integrated **MCP** tooling (**ThreatStream MCP**, Google Workspace **MCP** via langchain-mcp-adapters) to extend agent capabilities with live enrichment and productivity tools.
- Drove **ThreatStream** IOC enrichment work — batch enrichment of IPs, domains, and hashes, and built intelligence dashboards visualizing 100+ IOC records across categories.
- Contributed design work on the Agent Platform for Starship, **ThreatStream**'s next-generation UI, and produced PDF API reference documentation for the platform's Alert Triage and Incident Response APIs.

Dhruvo — Real-Time AI Voice Calling Pipeline

Project Duration: 2025 – Present

Dhruvo is a product I've contributed to at Brain Station 23; my core contribution has been R&D on its real-time AI calling pipeline, which connects an actual phone number to an AI agent so the AI can place and receive genuine phone calls rather than simulate voice inside a browser. Calls are routed through **Asterisk** over SIP for telephony signaling and carrier connectivity, while **LiveKit** handles the real-time WebRTC media stream between the live call and the AI agent, with an **LLM** reasoning over the conversation in-flight to decide what to say and when. The hardest engineering problems were latency and turn-taking — keeping the pipeline fast enough that a conversation over a real phone line feels natural rather than robotic or delayed.

Tech Stack: LLM · Asterisk · LiveKit · Python · WebSockets

- Led R&D on the telephony bridge connecting a real caller number to **Asterisk** (SIP) for inbound and outbound call routing.
- Integrated **LiveKit** for real-time WebRTC audio streaming between the live phone call and the AI agent.
- Designed the conversational reasoning loop so the **LLM** could handle natural turn-taking and interruptions mid-call.
- Tuned the pipeline end-to-end for low-latency, real-time voice response suitable for production phone conversations.

NeuraFlow — No-Code AI Agent Builder

Project Duration: 2025 – Present

NeuraFlow is a platform for building AI agents visually, by dragging and connecting nodes on a canvas instead of hand-coding agent orchestration logic. Once an agent is built, it isn't locked to a single surface — the same agent definition can be deployed as a web chat widget, or connected to **WhatsApp**, **Microsoft Teams**, **Slack**, or **SMS**, so a business can build an agent once and reach users everywhere they already are, without rebuilding the logic per channel.

Tech Stack: LangGraph · LangChain · React · Node.js · WhatsApp · Microsoft Teams · Slack · SMS

- Built a drag-and-drop visual builder for composing AI agent logic as a node graph, removing the need to hand-code orchestration for each new agent.
- Designed a channel-agnostic agent runtime so a single agent definition deploys unchanged across a web widget, **WhatsApp**, **Microsoft Teams**, **Slack**, and **SMS**.
- Implemented the multi-channel integration layer connecting the agent runtime to each platform's messaging API.

CAD

Project Duration: Jun 2025 – Sep 2025

A voice-driven assistant that lets users perform website activities entirely through natural language — creating events, filling out forms, and managing tasks across web platforms without touching a keyboard or mouse. The assistant listens continuously, interprets user intent through an **LLM**-driven reasoning layer, maps that intent to concrete UI actions, and executes them on the page in real time, giving users a hands-free way to operate web applications they'd otherwise have to click through manually.

Tech Stack: OpenAI · LangChain · Redis · Web Speech API · Node.js · React

- Designed and developed the voice assistant end-to-end, integrating an AI-driven workflow using **LangChain** and **OpenAI** to interpret natural language commands.
- Implemented real-time voice recognition and processing using the **Web Speech API**.
- Managed session and conversation state with **Redis** for reliable, low-latency interactions.
- Built the frontend interface with **React**, delivering seamless voice-guided navigation across web applications.

Nissan Drivers Guide – Qashqai 2025

Project Duration: Mar 2025 – Jul 2025 | poc.cmcge-converters.eu

A comprehensive digital owner's manual (OM) platform that puts the entire Nissan Qashqai 2025 manual inside a mobile app, replacing a dense paper document with an interactive, searchable experience. The core challenge was trust: the AI chatbot had to answer strictly from the actual manual content and never fabricate instructions for a physical vehicle, so it was built as a retrieval-grounded agent using **LangChain** and **LangGraph** with an explicit think-plan-execute loop rather than a free-form chatbot. Alongside the chatbot, users get keyword-based document search across the manual and an interactive **Three.js** 3D vehicle viewer for exploring features visually, all backed by a **FastAPI** and **PostgreSQL** service layer tuned for a smooth, low-latency in-app experience.

Tech Stack: AI · NLP · Three.js · FastAPI · PostgreSQL · AWS · LangChain · LangGraph · Redis

- Led the design and development of the digital OM platform end-to-end, from architecture through deployment.
- Built a retrieval-grounded AI/NLP chatbot over the full owner's manual using **LangChain** and **LangGraph**, giving the agent a robust think-plan-execute loop and ensuring answers stayed accurate and hallucination-free.
- Implemented advanced keyword-based search functionality so users could quickly locate the right manual section.
- Incorporated an interactive **Three.js** 3D vehicle viewer and optimized backend performance with **FastAPI** and **PostgreSQL** to ensure a seamless user experience.

UpendNow

Project Duration: Oct 2024 – Mar 2025 | upendnow.xyz

A platform that takes a content creator from a raw idea to a full pre-production package for a video ad or short film, without needing a production team to get started. From a short prompt, the system generates scripts, creative briefs, and storyboards with AI-generated concept imagery and suggested color palettes, giving creators in a single guided workflow what they'd normally spend days assembling by hand. Built on a serverless **AWS** architecture, it orchestrates **OpenAI** for scriptwriting and creative briefs, **Midjourney** for scene and concept art, and **Hugging Face's** Minimax models for AI-generated video sequences, tying them together into one coherent creative pipeline.

Tech Stack: OpenAI · Midjourney · Node.js · Serverless · AWS

- Spearheaded the architectural design and full-stack development of a scalable enterprise system using cloud-native services and a serverless, microservices architecture on **AWS Lambda**.
- Directed project lifecycle management for a cross-functional team of engineers, delivering core features ahead of schedule.
- Engineered robust API integrations embedding **OpenAI** and **Midjourney** AI-driven workflows, improving system efficiency by over 20%.

Nissan Drivers Guide Me

Project Duration: May 2024 – Oct 2024 | app.nmef.xyz

A management dashboard and backend built for an automobile company to organize and publish the digital assets that power their companion mobile app — AR content files, EPUB manuals, images, and PDFs — from a single admin interface instead of scattered file drops. The system needed to serve fairly large binary assets to the app quickly and consistently, so a large part of the work went into caching strategy: keeping frequently accessed assets fast for end users while keeping the admin-side upload and versioning workflow simple for the content team managing it.

Tech Stack: Node.js · Next.js · NestJS · PostgreSQL · Tailwind CSS

- Built the management dashboard and backend using **Next.js**, **NestJS**, and **PostgreSQL** as a full-stack developer.
- Developed asset-management workflows supporting AR content files, EPUB, images, and PDFs for in-app showcasing.
- Implemented efficient caching strategies to optimize system performance and ensure seamless content delivery.

Medilynq

Project Duration: Apr 2023 – Oct 2023 | app.medilynq.com

A web-based system built for clinics doing cardiovascular (ECG/Holter) monitoring, turning raw cardiac signal data into interpretable, shareable clinical reports. Medical professionals feed in patient monitoring data, run it through the platform's signal-analysis workflows, and generate structured diagnostic reports through an interface designed to fit a fast-paced clinical workflow rather than require specialist software training. The focus throughout was end-to-end turnaround: getting from raw signal to a report a physician can act on as quickly and reliably as possible.

Tech Stack: Node.js · MongoDB · React

- Spearheaded full-stack development of the platform using **Node.js**, **MongoDB**, and **React**.
- Built signal-analysis and report-generation workflows tailored to clinical use cases.
- Designed an intuitive interface enabling medical professionals to interpret cardio signals efficiently.

Fitsomnia

Project Duration: Dec 2022 – Apr 2023 | fitsomnia.com

A social fitness platform where the backend had to keep up with a growing volume of user-generated photo and video content without slowing down the rest of the app. As a backend developer, the core contribution was moving image and video processing off the request path entirely — into an asynchronous serverless pipeline using **Python** and **OpenCV** — so uploads returned instantly to the user while processing ran in the background, with Amazon **S3** handling storage at scale. That shift from synchronous to asynchronous processing was the direct driver of the platform's 30% performance improvement.

Tech Stack: OpenCV · AWS Lambda · Amazon S3

- Implemented a serverless Lambda function with **Python** to process images and videos asynchronously.

- Transitioned processing from a synchronous to an asynchronous backend architecture, improving performance by 30%.
- Used **OpenCV** for media processing and Amazon **S3** for scalable storage as a backend developer.

Cubenationshop

Project Duration: 2022 May – 2022 Dec | cubenationshop.com

An e-commerce platform built for CubeNation, one of Bangladesh's largest speedcubing and puzzle stores, designed to handle high traffic and fast page loads for a catalog of niche, collectible products. Beyond the customer-facing storefront, the project included a dedicated **React.js** admin panel giving the CubeNation team direct control over products, orders, and coupons without engineering involvement for routine changes, plus an **Elasticsearch**-backed search system so customers could find specific cubes and puzzles quickly across a large and growing catalog.

Tech Stack: **React.js** · **Next.js** · **Nest.js** · **MongoDB** · **Tailwind CSS** · **Elasticsearch**

- Contributed to designing the project infrastructure to be scalable and efficient for high-traffic loads.
- Developed a user-friendly frontend using **Next.js** and **Tailwind CSS**, plus a dedicated **React.js** admin panel.
- Implemented RESTful backend APIs and a fast, efficient **Elasticsearch**-based search system.
- Resolved critical business logic problems for the client and conducted code reviews to maintain quality standards.

Stayy

A platform for booking private accommodations through trusted friends and family rather than anonymous listings, built directly from requirements gathered in ongoing conversation with the client rather than a fixed spec handed off up front. The UI was built with **Next.js** and **Tailwind CSS**, authentication ran through **AWS Cognito**, and the backend was a fully RESTful **Nest.js** API; the whole system was deployed on **AWS (EC2, S3, Elastic Beanstalk)** with CI/CD pipelines through **CodePipeline**, **CodeBuild**, and **GitHub** workflows so changes could ship continuously as requirements evolved.

Tech Stack: **TypeScript** · **React** · **Next.js** · **Nest.js** · **Postgres** · **Material UI** · **Tailwind CSS** · **AWS Cognito** · **AWS S3** · **Amazon EC2** · **Elastic Beanstalk** · **CodePipeline** · **CodeBuild**

- Designed and built the UI using **Next.js** and **Tailwind CSS** based on direct client requirements.
- Implemented an authentication system using **AWS Cognito**.
- Built a fully RESTful API with **Nest.js**.
- Deployed on **AWS** infrastructure (**EC2** for virtual servers, **S3** for storage, **Elastic Beanstalk** for hosting) with CI/CD pipelines via **CodePipeline**, **CodeBuild**, and **GitHub** workflows.

AREAS OF INTEREST

Distributed Systems, System Design, Cloud Computing